1.FIND-S ALGORITHM

```
print("J THASLIMA NASREEN 192424152")
import pandas as pd
# Read CSV file
df = pd.read_csv("find s  - Sheet1.csv")
# Rename the last column
df.rename(columns={df.columns[-1]: "Allergy"}, inplace=True)
print("Training Data:\n")
print(df)
# Ignore the Number column
concepts = df.iloc[:, 1:-1].values
target = df.iloc[:, -1].values
# Initialize hypothesis
hypothesis = ['0'] * len(concepts[0])
print("\nInitial Hypothesis:", hypothesis)
for i in range(len(target)):
    if target[i] == "Yes":
        if hypothesis == ['0'] * len(concepts[0]):
            hypothesis = list(concepts[i])

        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != concepts[i][j]:
                    hypothesis[j] = '?'

        print("\nAfter Positive Example", i + 1)
        print(hypothesis)

print("\nFinal Hypothesis")
print(hypothesis)
```

OUTPUT

```
J THASLIMA NASREEN
(192424152)
Training Data:

   Number Restaurant       Meal       Day       Cost Allergy
0  Visit1      Sam's  Breakfast    Friday      Cheap     Yes
1  Visit2      Kim's      Lunch    Friday  Expensive      No
2  Visit3      Sam's      Lunch  Saturday      Cheap     Yes
3  Visit4      Bob's  Breakfast    Sunday      Cheap      No
4  Visit5      Sam's  Breakfast    Sunday  Expensive      No

Initial Hypothesis: ['0', '0', '0', '0']

After Positive Example 1
["Sam's", 'Breakfast', 'Friday', 'Cheap']

After Positive Example 3
["Sam's", '?', '?', 'Cheap']

Final Hypothesis
["Sam's", '?', '?', 'Cheap']
```